ChatGPT

Pathway Analysis and Enrichment

Modern genomics studies often produce long lists of differentially expressed (DE) genes. To interpret these genes, we ask: *what biological functions or pathways are affected?* **Pathway analysis** (or *gene set enrichment analysis*) tests whether predefined groups of genes (pathways, Gene Ontology categories, etc.) are over-represented among DE genes. This adds biological meaning beyond individual genes. For example, in cancer studies clustering gene expression profiles (hierarchical clustering) can reveal patterns distinguishing tumor subtypes pubmed.ncbi.nlm.nih.gov 【62†】, and enrichment analysis can identify if particular pathways underlie those clusters. In fact, the classic study by Golub *et al.* showed that unsupervised clustering of leukemia gene-expression data could automatically separate acute myeloid and lymphoblastic leukemia (without any prior labels) pubmed.ncbi.nlm.nih.gov. After clustering or other expression analysis, pathway/gene-set methods help connect gene-level changes to broader biological themes (e.g. "cell cycle" or "immune response").

Gene sets and databases: Common gene sets include curated pathways (e.g. **KEGG** pathways), Gene Ontology (GO) categories, and user-defined gene modules. The Gene Ontology is a hierarchical vocabulary of gene function organized as a directed acyclic graph (DAG) of terms. It spans three branches: *Biological Process*, *Molecular Function*, and *Cellular Component* nature.com. For example, a GO DAG might show "DNA repair" and its child terms. Databases like **KEGG** provide signaling and metabolic pathway maps. Collections such as the **MSigDB** (Molecular Signatures Database) curated by the Broad Institute collect many gene sets (e.g. canonical pathways, literature gene lists, cancer hallmarks) for human and model organisms. Tools for enrichment analysis rely on these resources to ask whether DE genes are enriched in any such set.

Over-Representation Analysis (ORA)

Over-representation analysis (ORA) is the simplest enrichment method. We start with a list of “interesting” genes (e.g. those with significant differential expression) and a larger background (“universe”) of all tested genes. For each gene set, we ask: *Is this set represented among the interesting genes more than expected by chance?* We can think of the classic urn model: the universe of N genes contains K genes belonging to the pathway (the “success” category) and N–K genes outside it. If we randomly draw n genes (the DE list) without replacement, the chance of drawing **k** or more pathway genes can be computed from the hypergeometric distribution. Intuitively, imagine an urn with K green balls (genes in the pathway) and (N–K) white balls (other genes). Drawing n balls without replacement, the probability of drawing exactly k green balls is

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}.$$

Enrichment is assessed by the probability of *k* or more successes (green balls) under this null model (i.e. the pathway is no more frequent than random). In R one can use `phyper()` or `fisher.test()`. For example:

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```r
# Example: N=20000 total genes, K=500 genes in a pathway, n=100 DE genes, k=8 ov
N <- 20000; K <- 500; n <- 100; k <- 8
# p-value for >=8 successes:
pval <- phyper(k-1, K, N-K, n, lower.tail=FALSE)
print(pval)
```

If p is small, the pathway is significantly enriched in the DE list. This is exactly Fisher's exact test on the 2×2 contingency table. ORA yields a p-value for each gene set, and these p-values are typically adjusted for multiple testing (e.g. Benjamini–Hochberg FDR).

Practically, ORA is implemented in tools like **DAVID** and R packages. For example, the Bioconductor package **clusterProfiler** provides functions like `enrichGO()`, `enrichKEGG()`, `enrichPathway()` etc., which take a vector of gene IDs and test for GO/KEGG over-representation. DAVID (Database for Annotation, Visualization and Integrated Discovery) is a popular web-based resource: it accepts a gene list (human, mouse, etc.) and annotates and tests for enrichment in GO categories, KEGG pathways, and other terms genomebiology.biomedcentral.com. DAVID organizes results into modules (e.g. GO charts, KEGG charts) for visual summaries genomebiology.biomedcentral.com. Other ORA tools include **WebGestalt**, **Enrichr**, **ShinyGO**, and **ReactomePA**. In all these, the core statistic is hypergeometric/Fisher's test, the same basic math applies (often referred to as the "gene set overlap test").

Gene Set Enrichment Analysis (GSEA) and Similar Methods

An alternative to ORA is GSEA-style enrichment, which uses the entire ranked gene list instead of a cutoff. Original GSEA (Subramanian *et al.*, 2005) computes an enrichment score by walking down the ranked list of all genes (ranked by expression fold-change or a test statistic) and increasing a running sum when a gene is in the gene set and decreasing it otherwise. The maximum deviation of this running-sum statistic (like a weighted Kolmogorov–Smirnov test) measures enrichment. Significance is assessed by permuting gene labels or sample labels. The key idea is that even if individual genes are not highly significant, a coordinated shift of many set members can yield a significant enrichment.

In R/Bioconductor, GSEA-like methods include `gseGO()` or `gseKEGG()` in **clusterProfiler**, and the **fgsea** package. The Broad Institute's GSEA software (with MSigDB gene sets) is widely used. For example, to run GSEA via clusterProfiler one might do:

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```
library(clusterProfiler)
# Suppose 'gene_list' is a named vector of gene log2FC ordered descending,
# with names as gene Entrez IDs.
gsea_res <- gseGO(geneList=gene_list, OrgDb=org.Hs.eg.db,
                  ont="BP", nPerm=1000)
```

or use `fgsea::fgsea()` on a *named* vector of statistics with gene sets from MSigDB (via `msigdbr`).

The difference between ORA and GSEA approaches: ORA collapses each gene to "in" or "out" of a list (losing magnitude/rank information and depending on an arbitrary cutoff), while GSEA uses the continuous ranking information and does not require a cutoff. ORA can miss pathways where many genes change moderately; GSEA can detect those patterns bmcbioinformatics.biomedcentral.com. Conversely, ORA is simpler and does not need permutation tests, making it faster and easier for smaller gene sets. In practice, both approaches can be complementary.

Statistical Basis: The Hypergeometric Test (Urn Model)

Mathematically, the hypergeometric distribution underlies ORA. If our background has N total genes, and a gene set of interest contains K genes, the probability of seeing exactly k of them among n sampled DE genes is given by the hypergeometric mass function above. To compute the *p*-value for enrichment, we sum probabilities of observing k or more in a random draw of size n. Conceptually, imagine an urn with K green balls (genes in the pathway) and $N - K$ white balls. Drawing n balls without replacement, asking "what is the probability of drawing at least k green balls?" This urn metaphor makes the dependence structure clear: draws are not independent (once a green is drawn, there are fewer remaining). If one instead drew with replacement, it would be a binomial model (used when sampling fraction is small). But for gene sets, hypergeometric without replacement is correct. In R, the `phyper()` function gives cumulative probabilities; for example `phyper(k-1, K, N-K, n, lower.tail=FALSE)` yields $P(X \geq k)$. Fisher's exact test on the 2×2 contingency table of in-vs-out of the list and gene set is equivalent to the one-sided hypergeometric test.

Hypergeometric *intuition:* suppose there are 10,000 genes, 200 of which are in pathway A. Among your 500 DE genes, 20 are in pathway A. Under random chance (0.02 of genes in A), you would expect ~10 (0.02×500) in A. Getting 20 or more by random is unlikely. The hypergeometric calculation or `phyper()` computes the exact probability. Visualization can help: often one draws a Venn diagram showing the overlap of DE genes and the pathway gene set (as two circles) to see how large the overlap is relative to each set carpentries-incubator.github.io . For example, if 24% of DE genes fall in the pathway versus only 5% of all genes, that suggests enrichment carpentries-incubator.github.io (see Figures in many tutorials).

Tools and Resources

Many online tools aggregate annotation and enrichment analysis. **DAVID** (NIH) provides an easy interface: paste a gene list, select identifier type, and it returns enriched terms (GO, KEGG, Pfam domains, etc.) with enrichment statistics genomebiology.biomedcentral.com. **GSEA software** (Broad Institute) can perform ranked-list enrichment using MSigDB gene sets; MSigDB itself catalogs thousands of human and mouse gene sets (e.g. C2 curated sets, C5 GO terms, H hallmark sets). **KEGG Pathway** database can be queried or visualized (e.g. via the R package `pathview` or `clusterProfiler::enrichKEGG`) to map genes onto metabolic/signal pathways. **Gene Ontology** consortium (geneontology.org) provides up-to-date GO term annotations; tools like `enrichGO()` in clusterProfiler tap this. **ShinyGO** is a user-friendly web application (Wang *et al.*, 2020) that performs GO enrichment and visualizes results with bar plots, network graphs of related GO terms, and KEGG pathway views. It also shows networks of enriched terms connected by gene overlap, and even retrieves STRING protein interaction networks for the gene list bioinformatics.sdstate.edu bioinformatics.sdstate.edu. (Other tools: **Enrichr**, **WebGestalt**, **Panther**, etc., offer similar functions.)

Visualization of Results

Enrichment results are often summarized in tables or bar plots of top GO/pathway terms by significance. Network-based visualizations can help: for instance, enriched pathways are nodes connected if they share genes (as in ShinyGO's network plot bioinformatics.sdstate.edu). A common portrayal is a *dotplot* or *barplot* of –log10(p) for top terms. Because ORA is inherently about set overlaps, simple Venn diagrams can illustrate the overlap between the DE list and a particular gene set carpentries-incubator.github.io. Heatmaps with clustering (as below) are used to visualize expression data clusters, which can then be annotated by enriched functions.

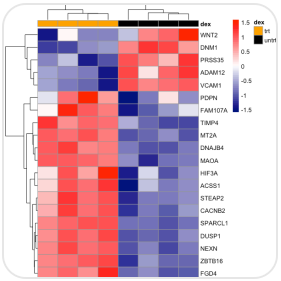


Figure: Example heatmap of gene expression with hierarchical clustering dendrograms on both rows (genes) and columns (samples). Clusters of co-expressed genes (rows) may correspond to shared biological functions; one can test these clusters for GO-term enrichment. (Image adapted from NIH tutorial【62†】.)

Integrating Clustering and Functional Annotation

Hierarchical clustering groups genes by expression similarity, revealing modules of co-regulated genes. We can then ask if these modules share functional themes: e.g. “Cluster 1 genes are enriched in cell cycle and DNA repair GO terms, suggesting this cluster reflects proliferative activity.” This two-step workflow is common: cluster expression profiles (or samples), then perform enrichment on each gene cluster. As illustrated by Golub *et al.* in cancer expression data, clustering can identify biologically meaningful groups of genes or samples pubmed.ncbi.nlm.nih.gov. Once clusters are defined, tools like `clusterProfiler::compareCluster()` can run enrichment on multiple gene lists (clusters) and summarize shared versus unique annotations. In summary, unsupervised clustering and enrichment together help distill large datasets into human-interpretable biology.

Key references: Golub *et al.* 1999 for clustering in cancer profiles pubmed.ncbi.nlm.nih.gov; Luo *et al.* 2009 describing the GAGE enrichment method bmcbioinformatics.biomedcentral.com; DAVID documentation by Dennis *et al.* 2003 genomebiology.biomedcentral.com; Gene Ontology Consortium 2000 nature.com; Wang *et al.* 2020 for ShinyGO. Other useful resources include the Bioconductor documentation for **clusterProfiler** (Yu *et al.* 2012) and **gage**, as well as the Broad GSEA guide and MSigDB help pages.

Citations

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